School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78AB

Actuarial and Financial Mathematics

SOLUTIONS

August 2024-2025

Duration: 2 Hours

Total marks available:

Question 1

(a) Solution

Annual effective i = 0.05

Monthly effective j = $(1.05)^{1/12} - 1 = 0.004074$ (1)

Monthly discount rate d = j/(1+j) (1)

= 0.004074/1.004074 = 0.004058 = 0.4058% (1)

[3 marks]

(b) (i) Solution

$$a_{\overline{2n|}} = \ddot{a}_{\overline{2n+1|}} - 1 = 23 - 1 = 22$$ (1)

[1 mark]

(ii) Solution

$$a_{\overline{2n|}} = a_{\overline{n|}} + v^n a_{\overline{n|}}$$ (2)

$$22 = 13 + 13v^n$$
$$v^n = 0.69231$$ (1)

[3 marks]

Total [7 marks]

Question 2

a) Solution:

i = 0.065/12 = 0.0054166667 (1)

315,000 = $Xa_{300|i}$

Calculate X = 2,126.90 (1)

[2 marks]

b) Solution:

New payment amount, X* = 2,126.90 + 125 = 2,251.90 (1)

Let n be the number of payments to be made.

$315{,}000 = 2251.90\ a_{n|i}$ (1)

Calculate n = 262.4094 (1)

Julia will make 262 full payments and 1 partial payment, or 263 payments. (1)

[4 marks]

c) Solution:

$OB_{262} = 315000(1 + i)^{262} - 2251.902558\ s_{262|i}$ (2)

$= 918.3311$ (1)

Last payment = 918.3311(1 + i) = 923.31 (1)

[4 marks]

Total 10 marks

Question 3

(a) Solution:

NPV = 0

$2 = v + 3v^2$ (1)

$3v^2 + v - 2 = 0$

$(3v - 2)(v + 1) = 0$

$v = 2/3$ ($v = -1$ is not practical) (1)

Solve i = 0.5 (1)

Marks also awarded if a numerical method is used, leading to the correct answer.

[3 marks]

(b) Solution:

Any three of the following ((1) for each correct answer):

- There may be limited capital and another potential project has a higher NPV
- The discount rate used may not have considered the full risk of the project, and would have been too low
- The payback period may be too long

- The project may not meet the company's accepted standards for investment (e.g. green investing)
- The company may need to use its capital to resolve debt issues
- The tax considerations make it undesirable at the particular time
- Any other sensible answer

Max 3 marks

Part C solution

(i)

Solution:

Date	Before Deposit	After Deposit
1/1/22	500,000	
30/6/22	540,000	560,000
31/12/22	550,000	570,000
30/6/23	580,000	600,000
31/12/23	620,000	640,000
30/6/24	650,000	670,000
31/12/24	710,000	

(1)

Converting the annual rate to a semi-annual rate:

$(1+i)^2 = 1+j$

$(1+i)^2 = 1+ 0.06285$

Semi-annual i = 0.03095 (1)

Working in half-years:

$500(1+ i)^6 + 20(1+ i)^5 +20(1+ i)^4 + 20(1+ i)^3+ 20(1+ i)^2 + 20(1+ i)^1 = 710$ (1)

Test i = 0.03095 to verify the solution to the equation (1)

[4 marks]

(ii) TWRR

$(1+TWRR)^3 = 540/500 * 550/520 * 580/570 * 620/600 * 650/640 * 710/670 = 1.200$ (2)

TWRR = 0.062763. (1)

[3 marks]

(iii)

TWRR is a better measure for comparing performance of managers because it is not affected by size and timing of cashflows.

1 mark for sensible comment

[Total 14 marks]

Question 4

PV = $20{,}000v^5 [1 + (1.035)v + (1.035)^2 v^2 + (1.025)^3 v^3 + \ldots\ldots + (1.025)^{29} v^{29}]$ (1)

= $20{,}000\ v^5_{.06}$ {a_due_30] at rate *j,* where $1/1+j = 1.035/1.06$, therefore $j = 0.024155$ (1)

= $20{,}000\ v^5_{.06} (1 - v_j^{30}) / d_j$ where $d = 0.0241555 / 1.0241555 = 0.023585$ (1)

PV = £324,003.56 (1)

Alternatively: $\frac{20000 \times 1.06^{-4}}{1.035} \times a_{\overline{30}|}^{\ j}$

Total 4 marks

Question 5

Marks – a) and b) 1 for correct statement of the value, one for correct answer, c) 1 mark [Total 5 marks]

(a) $e^{\int_0^6 (.06 + .002t)\,dt}$

$= e^{\left[.06t + \frac{.002}{2}t^2\right]_0^6}$

$= e^{\left[.06 \times 6 + \frac{.002}{2}(36)\right]}$

$= e^{0.396}$

$= 1.485869$ is the accumulated value at $t = 6$

(b) $e^{\int_0^7 (.06 + .002t)\,dt}$

$= e^{\left[.06t + \frac{.002}{2}t^2\right]_0^7}$

$= e^{\left(.06 \times 7 + \frac{.002}{2}(49)\right)}$

$= e^{0.469}$

$= 1.598395$ is the accumulated value at $t = 7$

(c) Accumulated value from time 6 to 7

$= \frac{(b)}{(a)} = \frac{1.598395}{1.485869}$

$= 1.0757308$

Question 6

Marks as indicated [ ]
Total 6 + 2 = 8 marks

(a) First check if there will be a capital gain,

i.e. is $i > (1-t)g$

$8 > (1-0.2)7$ [1]

$\Rightarrow$ there will be a capital gain

Let P = the price per £100 nominal [2]

$$= \left[7a^{8\%}_{\overline{10}|} \times 0.8 + 100v^{10}_{8\%} - 0.2(100-P)v^{10}_{8\%}\right] \times (1.08)^{-1/2}$$

where $a^{8\%}_{\overline{10}|} = \frac{1 - v^{10}_{.08}}{0.08} = 6.710081$ [1]

$$v^{10}_{8\%} = 0.463193$$

$$P = \left[37.576456 + 46.3193 - 9.2638698 + 0.2(0.463193)P\right](1.08)^{-1/2}$$

$$= \left(74.631935 + 0.0926387P\right)(1.08)^{-1/2}$$

$$0.96225P = 74.631935 + 0.0926387P$$

$$0.8696117P = 74.61395$$

$$P = £85.80$$ [2]

(b) Let e be expected inflation

$(1+e)(1.025) = 1.08$ [1]

$\rightarrow e = 5.366\%$ pa annum. [1]

Question 7

a) and b) Marks as indicated [ ]

(a) Duration $= \dfrac{\sum_{t=1}^{n} t C_t v^t}{\sum C_t v^t}$ [1]

$= \dfrac{1xv + 2xv^2 + 3xv^3 + \ldots + nv^n}{v+v^2+v^3+\ldots+v^n}$ [1]

$= \dfrac{(Ia)_{\overline{n}|}}{a_{\overline{n}|}}$ [1]

Total 3 marks

(b) Liabilities $= a_{\overline{20}|} + a_{\overline{40}|}$

Duration $= \dfrac{(Ia)_{\overline{20}|} + (Ia)_{\overline{40}|}}{(2+v^{20})\, a_{\overline{20}|}}$ [1]

where $(Ia)_{\overline{20}|} = \dfrac{\ddot{a}_{\overline{20}|} - 20v^{20}}{i} = 125.155002$ [½]

$(Ia)_{\overline{40}|} = \dfrac{\ddot{a}_{\overline{40}|} - 40v^{40}}{i} = 306.323077$ [½]

$a_{\overline{20}|} = \dfrac{1-v^{20}}{i} = 13.590326$ [½]

$\ddot{a}_{\overline{20}|} = \dfrac{1-v^{20}}{d} = 14.133939$

$\ddot{a}_{\overline{40}|} = \dfrac{1-v^{40}}{d} = 20.584455$

Duration $= \dfrac{125.155002 + 306.323077}{(2+v^{20})\ 13.590326}$ [1]

$= \dfrac{431.478079}{33.383099}$ [½]

$= 12.925$ years

Total 4 marks

C) 1 for expression of duration, 1 for working, 1 for correct answer

d) 2 marks for sensible comment

Total 12 marks

(b) Duration of the bonds

$$= \frac{\sum_{t=1}^{15} 8tv^t + 15 \times 100 v^{15}}{\sum_{t=1}^{15} 8v^t + 100 v^{15}} \qquad \text{all at } 4\%$$

$$= \frac{8(Ia)_{\overline{15}|} + 1500 v^{15}}{8a_{\overline{15}|} + 100 v^{15}}$$

$$a_{\overline{15}|} = \frac{1 - v^{15}}{i} = 11.118\,38744$$

$$(Ia)_{\overline{15}|} = \frac{\ddot{a}_{\overline{15}|} - 15v^{15}}{i} = 80.8538851$$

$$\ddot{a}_{\overline{15}|} = \frac{1 - v^{15}}{d} = 11.563122\,74$$

$$\text{Duration} = \frac{8 \times 80.8538851 + 1500 v^{15}}{8 \times 11.118\,38744 + 100 v^{15}}$$

$$= \frac{1479.727834}{88.9471 + 55.52645}$$

$$= 10.2422 \text{ years}$$

(c) For immunisation, the duration of the assets must equal the duration of the liabilities. It is not possible to purchase government bonds with a longer term therfore not possible to immunise the liabilities.

Notes for External Examiner

Question 1 *This question covers the topics of Interest Rates, PV, FV, and annuities*.

IFoA syllabus a) 1.1.2, b) 1.6

Question 2 *This question covers the topics of loans and loan schedules*.

IFoA syllabus 2.2.1

Question 3 *This question covers the topics of yields and discounted cash flows*.

IFoA syllabus 2.3

Question 4 This question covers the topics of annuities, deferred and compound increasing

IFoA syllabus 1.3, 1.5.3

Question 5 This question covers the topics of continuous cashflows and force of interest

IFoA syllabus 1.5.2

Question 6 This question covers the topics of bonds and tax

IFoA syllabus 2.2.2

Question 7 This question covers the topics of immunisation

IFoA syllabus 1.8